EPIGRAM

4-Week Intensive Quant Interview Prep

Week 1, Topic 1

Foundations of Probability Modeling

Counting · Inclusion-Exclusion · Geometric Probability · Combinatorics

Problems:	6 free + 7 premium
Estimated time:	90–120 minutes
Prerequisites:	None

Box types: **Worked** — fully solved. **[FREE]** — all users. **[PREMIUM]** — subscribers only.

Difficulty: **Easy** **Medium** **Hard** **Very Hard**

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By the end of this topic, you will be able to...

1. **Set up any probability problem** from scratch — define the sample space Ω, identify the event A, and compute $\mathbb{P}(A)$ for uniform probability spaces.
2. **Apply complementary counting** to handle "at least one" problems, including the Birthday Paradox and its generalisation.
3. **Use the Inclusion-Exclusion Principle** to compute probabilities involving overlapping events and consecutive patterns.
4. **Count arrangements under constraints** — circular permutations, non-adjacent selections, and the block trick for consecutive groups.
5. **Derive and apply the derangement formula**, and know that $\mathbb{P}(\text{derangement}) \approx 1/e$ for $n \geq 5$.
6. **Solve geometric probability problems** by mapping events onto regions and computing areas — including chord intersection and semicircle containment.
7. **Recognise and exploit symmetry** as the fastest path to an answer in interview settings.

*Problems 1–6 ([**FREE**]) build all of the above skills. Problems 7–11 and A1–A2 ([**PREMIUM**]) are the level you need to crack final rounds at Citadel, Jane Street, and Two Sigma.*

Introduction: What Is Probability?

Probability is the mathematics of uncertainty. In quantitative finance and trading, every model, every bet, and every risk estimate rests on probabilistic foundations. The goal of this topic is to build the *problem-solving toolkit* that lets you crack interview questions under pressure in 5–10 minutes.

The Three Ingredients of Any Probability Problem

1. **Sample space** Ω: the set of all possible outcomes.
2. **Event** $A \subseteq \Omega$: outcomes you care about.
3. **Probability measure** $\mathbb{P}$: $\mathbb{P}(\Omega) = 1$, $\mathbb{P}(\emptyset) = 0$, countable additivity.

For **uniform** spaces: $\mathbb{P}(A) = |A|/|\Omega| = \text{favourable/total}$.

Interview strategy: (1) Identify Ω. (2) Count $|A|$ — directly or via complementary counting / IEP. (3) Sanity-check boundary cases.

Counting Fundamentals

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Permutations and Combinations

Key Formulas

$$\text{Permutations (ordered):} \quad P(n,k) = \frac{n!}{(n-k)!}$$

$$\text{Combinations (unordered):} \quad \binom{n}{k} = \frac{n!}{k!\,(n-k)!}$$

Order matters ⇒ permutation. Order doesn't ⇒ combination.

Warm-Up

Worked Example 1 [FREE] ■■■■ Easy

Majority Heads in Three Coin Tosses. Flip a fair coin three times. What is the probability of getting more heads than tails?

Solution

$|\Omega| = 8$. Favourable ($\#\text{H} \geq 2$): {HHH, HHT, HTH, THH} = 4 outcomes.

$$\mathbb{P} = \frac{4}{8} = \frac{1}{2}$$

Slick: Odd tosses ⇒ no ties ⇒ by symmetry $\mathbb{P}(\#H > \#T) = \mathbb{P}(\#T > \#H) = \frac{1}{2}$.
Key takeaway: For any odd number of fair coin tosses, majority heads has probability $\frac{1}{2}$ by symmetry — no calculation needed.

Problem 2 ■■■■ Medium

Parity of Heads in Coin Tosses. Flip 10 fair coins. What is the probability the number of heads is even?
Hint: try the algebraic trick $(p + (1-p))^n \pm ((1-p) - p)^n$.

Solution

Method 1 — Direct: $|A| = \binom{10}{0} + \binom{10}{2} + \cdots + \binom{10}{10} = 512$, $\mathbb{P} = 512/1024 = \frac{1}{2}$.
Method 2 — Algebraic trick: $\mathbb{P}(\text{even heads}) = \left.\frac{1 + (1-2p)^n}{2}\right|_{p=1/2} = \frac{1}{2}$ for all $n \geq 1$.
Method 3 — Recurrence: $a_n = \frac{1}{2}$ satisfies $a_n = p(1 - a_{n-1}) + (1-p)a_{n-1}$ when $a_{n-1} = \frac{1}{2}$.
Key takeaway: Parity of heads is always $\frac{1}{2}$ for any $n \geq 1$ fair coin tosses.

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Complementary Counting

Technique: Complementary Counting

$\mathbb{P}(A) = 1 - \mathbb{P}(A^c)$. Use when "at least one" appears and the complement ("none" / "all different") is easier to count.

Problem 3 ■■■■ **Medium**

The Birthday Paradox. Assume 365 equally likely birthdays. Find the smallest n such that $\mathbb{P}(\text{at least two people share a birthday}) > \frac{1}{2}$.

Solution

$\mathbb{P}(\text{all distinct}) = \prod_{k=0}^{n-1}(1 - k/365)$. Approximation $\ln(1-x) \approx -x$: threshold at $n(n-1) > 730 \ln 2 \approx 506$, so $\mathbf{n = 23}$.

Key takeaway: Pairs grow as $\binom{n}{2}$, reaching ~ 365 when $n \approx 23$. General threshold: $n \approx 1.18\sqrt{m}$ for pool size m.

Combinatorial Techniques

Technique: Non-Adjacent Selection

Choose k from $\{1, \ldots, n\}$ with no two adjacent: $\binom{n-k+1}{k}$ ways. (Substitute $y_i = x_i - (i-1)$ to absorb gap constraints.)

Problem 4 ■■■■ **Medium**

Probability of Choosing Three Non-Consecutive Integers. From $\{1, \ldots, 10\}$, pick 3 distinct numbers uniformly at random. What is the probability no two are adjacent?

Solution

Total: $\binom{10}{3} = 120$. Favourable: $\binom{8}{3} = 56$. $\mathbb{P} = 56/120 = \mathbf{7/15}$.

Key takeaway: Memorise $\binom{n-k+1}{k}$ — it turns a constrained adjacency problem into an unconstrained one.

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Inclusion-Exclusion Principle

Inclusion-Exclusion Principle (IEP)

$$\mathbb{P}\Big(\bigcup_{i=1}^{n} A_i\Big) = \sum_i \mathbb{P}(A_i) - \sum_{i<j} \mathbb{P}(A_i \cap A_j) + \sum_{i<j<k} \mathbb{P}(A_i \cap A_j \cap A_k) - \cdots$$

Signal phrase: **"at least one"**.

Problem 5 ■■■■ **Medium**

Probability of Consecutive Sixes in Three Dice Rolls. Roll a fair six-sided die 3 times. What is the probability of getting at least one occurrence of the consecutive pattern 66?

Solution

Let $A = \{\text{rolls 1,2 both 6}\}$, $B = \{\text{rolls 2,3 both 6}\}$.

$$\mathbb{P}(A \cup B) = \frac{1}{36} + \frac{1}{36} - \frac{1}{216} = \frac{11}{216} \approx 5.1\%$$

Key takeaway: Define $A_i = \{\text{pattern starts at } i\}$, apply IEP. Intersection $A_i \cap A_{i+1}$ is the longer overlapping pattern.

Circular Arrangements

Circular Permutations

Fix one person: $(n-1)!$ total. Block of k friends: $(n-k)! \cdot k!$ arrangements.

$$\mathbb{P}(k \text{ friends consecutive}) = \frac{k!\,(n-k)!}{(n-1)!}$$

Problem 6 ■■■■ **Medium**

Adjacent Friends Around a Round Table. **Part 1.** 5 people, 2 friends. $\mathbb{P}(\text{friends adjacent})$? **Part 2.** Generalise to n people, k friends all consecutive.

Solution

Part 1. Fix one friend; 4 positions, 2 adjacent: $\mathbb{P} = \mathbf{1/2}$.
Part 2. $\mathbb{P} = \dfrac{k!\,(n-k)!}{(n-1)!}$. Verify $n = 5, k = 2$: $12/24 = 1/2$. ✓
Key takeaway: Always fix one person first. The block trick handles all consecutive-group

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problems.

Advanced Counting and Geometric Probability

Card Dealing

Problem 7 ■■■■ **Hard** [PREMIUM]

Probability Each Player Receives an Ace. A 52-card deck is dealt to 4 players (13 cards each). What is the probability every player receives at least one ace?

Solution

Step 1: Total number of equally likely deals. A deal assigns 13 cards to each of 4 labeled players:

$$|\Omega| = \binom{52}{13}\binom{39}{13}\binom{26}{13}\binom{13}{13} = \frac{52!}{(13!)^4}$$

Step 2: Each player must receive exactly 1 ace. There is no other way to give all 4 players at least one ace (only 4 aces exist). The number of ways to assign the 4 distinct aces to the 4 labeled players is:

$$4!$$

Step 3: Deal the remaining 48 non-ace cards. Each player already has 1 ace and needs 12 more cards from the 48 non-aces:

$$\binom{48}{12}\binom{36}{12}\binom{24}{12}\binom{12}{12} = \frac{48!}{(12!)^4}$$

Step 4: Favorable deals.

$$|\text{favorable}| = 4! \cdot \frac{48!}{(12!)^4}$$

Step 5: Compute the probability.

$$\mathbb{P}(\text{each player has an ace}) = \frac{4! \cdot \dfrac{48!}{(12!)^4}}{\dfrac{52!}{(13!)^4}} = 4! \cdot \frac{48!}{52!} \cdot \frac{(13!)^4}{(12!)^4}$$

Step 6: Simplify.

$$\frac{48!}{52!} = \frac{1}{52 \cdot 51 \cdot 50 \cdot 49}, \qquad \left(\frac{13!}{12!}\right)^4 = 13^4$$

$$\mathbb{P}(\text{each player has an ace}) = \frac{4! \cdot 13^4}{52 \cdot 51 \cdot 50 \cdot 49}$$

Using $52 = 4 \cdot 13$ to cancel with $4! = 24$:

$$= \frac{24 \cdot 13^4}{4 \cdot 13 \cdot 51 \cdot 50 \cdot 49} = \frac{6 \cdot 13^3}{51 \cdot 50 \cdot 49}$$

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$$\boxed{\mathbb{P}(\text{each player has an ace}) = \frac{24 \cdot 13^4}{52 \cdot 51 \cdot 50 \cdot 49} = \frac{6 \cdot 13^3}{51 \cdot 50 \cdot 49} = \frac{2197}{20825} \approx 10.5\%}$$

Key takeaway: The direct-counting approach — assign aces first, then deal non-aces — is cleaner than IEP here. Despite 4 aces for 4 players, the probability each receives one is only ≈ 1 in 9.5, because most deals concentrate two or more aces in the same hand.

Derangements

Derangement Formula

$D_n = n! \sum_{k=0}^{n} (-1)^k / k! \approx n!/e$. For $n \geq 5$: $\mathbb{P}(\text{derangement}) \approx 1/e \approx 36.8\%$.

Problem 8

Medium [PREMIUM]

The Derangement Problem. 5 firms, 5 cover letters placed randomly. Probability no letter goes to the correct firm?

Solution

$$\mathbb{P} = \sum_{k=0}^{5} \frac{(-1)^k}{k!} = 1 - 1 + \tfrac{1}{2} - \tfrac{1}{6} + \tfrac{1}{24} - \tfrac{1}{120} = \frac{44}{120} = \frac{11}{30} \approx 36.7\%$$

Key takeaway: $\approx 1/e$ for $n \geq 5$. Recurrence: $D_n = (n-1)(D_{n-1} + D_{n-2})$.

Geometric Probability

Technique: Geometric Probability

$\mathbb{P}(A) =$ measure of favourable region/total measure. Identify the right sample space, map the event geometrically, then compute. **Always draw the picture.**

Problem 9

Hard [PREMIUM]

Probability All Random Points Lie in a Semicircle. Pick N points uniformly on a circle's circumference. What is the probability all N lie within some semicircle?

Solution

Let $A_i = \{\text{all other } N-1 \text{ points in the semicircle clockwise from } i\}$. Events are mutually exclusive, $\mathbb{P}(A_i) = (1/2)^{N-1}$.

$$\boxed{\mathbb{P} = \frac{N}{2^{N-1}}}$$

Key takeaway: Mutual exclusivity of $A_1, \ldots, A_N$ is the key step.

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Problem 10 Hard [PREMIUM]

Probability Two Random Chords Intersect. Choose chords by picking four endpoints uniformly on a circle. What is the probability the chords intersect?

Solution

4 random points $\to$ 3 equally likely pairings. Chords intersect iff endpoints alternate; exactly 1 of 3 pairings does this.

$$\boxed{\mathbb{P} = \frac{1}{3}}$$

Common Mistake

Different chord-sampling methods give different answers (Bertrand's paradox). Always clarify the sampling method in an interview.

Problem 11 Very Hard [PREMIUM]

Triangle Formation from Random Stick Breaks. **(1)** Unit stick cut at two uniform points. $\mathbb{P}$(triangle)? **(2)** First cut uniform; break the longer piece at a uniform point. $\mathbb{P}$(triangle)?

Solution

(1) Event = each piece $< 1/2$ = two triangles in $[0,1]^2$, total area $1/4$. $\boxed{\mathbb{P}_1 = 1/4}$.
(2) Condition on first cut $U \sim \text{Unif}(0, \frac{1}{2})$, second $V \sim \text{Unif}(U, 1)$. Need $V > 1/2$ and $V < U + 1/2$:

$$\mathbb{P}_2 = 2\int_0^{1/2} \frac{U}{1-U}\, dU = 2\ln 2 - 1 \approx 38.6\%$$

Key takeaway: Breaking the longer piece raises the probability from 25% to ≈38.6%.

Additional Exercises

Problem A1 Hard [PREMIUM]

Average Chord Length of the Unit Circle. What is the expected distance between two randomly chosen points on a unit circle?
Hint: chord length $= 2|\sin(\Delta\theta/2)|$*; integrate over* $\Delta\theta \sim \text{Unif}(0, \pi)$.

Problem A2 Medium [PREMIUM]

Balanced Assignment of Seniors Across Teams. 16 people (4 seniors, 12 juniors) partitioned into 4 labeled groups of 4. $\mathbb{P}$(each group has exactly 1 senior)?

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Hint: count ways to assign 4 seniors to 4 groups, one per group.

Technique Summary

Topic 1 Technique Toolkit

Uniform probability	Count favourable / count total.
Complementary	$\mathbb{P}(A) = 1 - \mathbb{P}(A^c)$. Use for "at least one".
Symmetry	Fair coin/dice: use symmetry before computing.
IEP	$\mathbb{P}(\bigcup A_i) = \sum \mathbb{P}(A_i) - \sum \mathbb{P}(A_i \cap A_j) + \cdots$
Non-adjacent	k from n, no two adjacent: $\binom{n-k+1}{k}$.
Circular	Fix one; block of k: $(n-k)! \cdot k!$.
Derangement	$\mathbb{P} \approx 1/e \approx 36.8\%$ for $n \geq 5$.
Geometric	Map to region; compute area. Draw it.
Chord intersection	4 pts $\to$ 3 pairings, 1 crosses: $\mathbb{P} = 1/3$.
Semicircle containment	$\mathbb{P} = N/2^{N-1}$.

Problem Index

#	Problem	Tier	Difficulty	Key Technique
W1	Majority Heads (3 coin tosses)	[FREE]	Easy	Symmetry
2	Parity of Heads (10 coins)	[FREE]	Medium	Algebraic trick
3	The Birthday Paradox	[FREE]	Medium	Complementary counting
4	Three Non-Consecutive Integers	[FREE]	Medium	Non-adjacent bijection
5	Consecutive Sixes in Dice Rolls	[FREE]	Medium	IEP
6	Adjacent Friends at Round Table	[FREE]	Medium	Circular permutations
7	Each Player Receives an Ace	[PREMIUM]	Hard	IEP, card counting
8	The Derangement Problem	[PREMIUM]	Medium	Derangement formula
9	All Points in a Semicircle	[PREMIUM]	Hard	Mutual exclusivity
10	Two Random Chords Intersect	[PREMIUM]	Hard	Symmetry, 3 pairings
11	Triangle from Stick Breaks	[PREMIUM]	Very Hard	Geometric probability
A1	Average Chord Length	[PREMIUM]	Hard	Geometric probability
A2	Balanced Senior Assignment	[PREMIUM]	Medium	Combinatorial counting

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